Hot Plate



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hot plate/stirrer

What it is used for

To heat laboratory samples or solutions using a hot plate, with stirring when a magnetic stirrer is fitted. To control heating and mixing while preventing burns, spills, and electrical hazards.

Main parts

- Heating surface or ceramic plate
- Temperature control
- Power switch and indicator
- Magnetic stirrer motor, if fitted
- Stir bar and sample vessel
- Support stand or probe connection, if fitted

Before you start

1. Read the instrument manual and confirm the vessel is suitable for heating.
2. Inspect the power cord, plug, heating surface, controls, and casing for damage.
3. Place the unit on a stable, dry, heat-resistant surface away from flammable materials.
4. Check that the sample vessel is intact and suitable for the expected temperature.
5. Add a compatible magnetic stir bar if stirring is required and position the vessel centrally.
6. Confirm that the temperature probe, if used, is suitable and correctly connected.

How to use

1. Ensure the controls are set to off before connecting the unit to power.
2. Add the sample to the vessel without overfilling it.
3. Place the vessel centrally on the heating surface and keep the area around it clear.
4. Connect power and switch on the unit according to the manual.
5. Set a low initial heating level and increase it gradually to the required setting.
6. If stirring is needed, start at a low speed and increase until mixing is steady without splashing.
7. Monitor the sample temperature using the approved sensor or thermometer when required.
8. Watch for boiling, foaming, bumping, or unexpected changes and reduce heat if needed.
9. Avoid touching the hot plate, vessel, or probe with bare hands during operation.
10. After heating, switch off the heater and stirrer and allow the vessel and surface to cool.
11. Disconnect power before cleaning or moving the instrument, once it is safe to do so.
12. Clean spills only after the unit has cooled and follow the laboratory's approved cleaning procedure.

⚠ Safety

- Do not heat sealed containers or vessels that are not rated for the operating temperature.
- Keep flammable solvents and vapours away unless the equipment is specifically approved for that application.
- Use suitable heat-resistant gloves, eye protection, and laboratory clothing.
- Assume the plate remains hot after switching off; do not touch until cooled.
- Keep liquids away from electrical controls and connectors.
- Never leave a hazardous heating process unattended and follow the laboratory's emergency procedures.

Care & storage

- Clean the surface after it cools using manufacturer-approved materials.
- Inspect the power cord, housing, controls, and probe for damage regularly.
- Keep ventilation openings clear and avoid abrasive tools that can damage the surface.
- Check temperature control and stirring operation periodically.
- Store stir bars and accessories clean and dry.

Common mistakes to avoid

- Setting the heater too high at the start and causing sudden boiling or splashing.
- Using a vessel that is unsuitable for the temperature or chemical.
- Touching the hot plate immediately after switching it off.
- Allowing spills to enter electrical controls or ventilation openings.
- Cleaning or moving the unit while it is hot or still connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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